

LECTURE 12

GROUP AND RING THEORY

9. FINITELY GENERATED MODULES OVER A PID

Here we prove the fundamental structure theorem for finitely generated modules over a principal ideal domain. This theorem provides a decomposition of such a module into a direct sum of cyclic modules. In particular, it gives an independent proof of the fundamental theorem of finitely generated abelian groups.

9.1. Decomposition theorem.

Theorem 9.1. *Let R be a principal ideal domain, and let M be any finitely generated R -module. Then*

$$M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r$$

is a direct sum of cyclic modules, where $s, r \in \mathbb{N} \cup \{0\}$, the a_i are non-zero non-units and $a_i \mid a_{i+1}$, for $i = 1, \dots, r-1$.

Proof. Since M is a finitely generated R -module, we have $M \cong R^n/K$ for some submodule K of R^n (cf. Lemma 5.24). Further by Lemma 5.25 we have $K \cong R^m$ for $m \leq n$. Let ϕ be this isomorphism from R^m to K . Thus $K = \phi(R^m)$. Let $\{e_1, \dots, e_m\}$ be a basis of R^m . Let us write

$$\phi(e_1) = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{n1} \end{pmatrix} \in R^n, \quad \dots, \quad \phi(e_m) = \begin{pmatrix} a_{1m} \\ a_{2m} \\ \vdots \\ a_{nm} \end{pmatrix} \in R^n.$$

Then $\phi(R^m) = AR^m$, where $A = (a_{ij})$ is an $n \times m$ matrix.

From Theorem 8.8, we can choose invertible matrices P and Q of order $n \times n$ and $m \times m$ respectively, such that $PAQ = \text{diag}(a_1, a_2, \dots, a_k, 0, \dots, 0)$, where $a_1 \mid a_2 \mid \cdots \mid a_k$. Then

$$M \cong \frac{R^n}{K} = \frac{R^n}{\phi(R^m)} = \frac{R^n}{AR^m} \cong \frac{R^n}{PAQR^m}$$

where the final isomorphism arises from composing ϕ with the respective change of basis transformations $\theta : R^m \rightarrow QR^m$ and $\zeta : R^n \rightarrow PR^n$, i.e. $\phi \mapsto \zeta\phi\theta$. So

$$\begin{aligned} M &= R^n / \left(\begin{array}{ccccccc} a_1 & & & & & & \\ & a_2 & & & & & \\ & & \ddots & & & & \\ & & & a_k & & & \\ & & & & 0 & & \\ & & & & & \ddots & \\ & & & & & & 0 \end{array} \right) \left(\begin{array}{c} R \\ R \\ \vdots \\ R \end{array} \right) \\ &= \frac{R \oplus R \oplus \cdots \oplus R}{Ra_1 \oplus Ra_2 \oplus \cdots \oplus Ra_k} \\ &\cong \frac{R}{Ra_1} \oplus \frac{R}{Ra_2} \oplus \cdots \oplus \frac{R}{Ra_k} \oplus R \oplus \cdots \oplus R \quad (\text{by Theorem 4.33}). \end{aligned}$$

Then, by deleting the zero terms if any - corresponding to those a_i 's that are units, and writing $s := n - k$, we have for some $u \in \{1, \dots, r\}$ that

$$\begin{aligned} M &\cong \frac{R}{Ra_u} \oplus \cdots \oplus \frac{R}{Ra_k} \oplus R^s \\ &= \frac{R}{Ra_1} \oplus \cdots \oplus \frac{R}{Ra_r} \oplus R^s \quad (\text{by renumbering if necessary}). \end{aligned}$$

As for any ideal I (including (0)), we have R/I is a cyclic R -module, the proof follows. $\square$

9.2. Uniqueness of the decomposition. First, we need some definitions.

Definition 9.2. Let R be a principal ideal domain. An element x of an R -module M is called a *torsion element* if there exists a non-zero element $r \in R$ such that $rx = 0$.

Example 9.3. Let $M = \mathbb{Z}/n\mathbb{Z}$ be a $\mathbb{Z}$ -module. Then, for all non-zero $x \in M$ we have $nx = 0$.

Definition 9.4. Let R be a principal ideal domain. A non-zero element x of an R -module M is called a *torsion-free element* if $rx = 0$ for $r \in R$ implies $r = 0$.

Example 9.5. Consider $R = \mathbb{Z}$ as a $\mathbb{Z}$ -module. Then, for all non-zero $x \in M$ we have $rx = 0$ if and only if $r = 0$.

The above two definitions can be extended to general rings R , but one needs to replace the restriction of r being non-zero in Definition 9.2 by not being a left or right zero divisor.

Theorem 9.6. Let R be a principal ideal domain, and let M be an R -module. Then

$$\text{Tor } M := \{x \in M \mid x \text{ is torsion}\}$$

is a submodule of M .

The proof is left as an exercise (see Exercise Sheet 13).

Theorem 9.7. Let M be a finitely generated module over a principal ideal domain R . Then

$$M = F \oplus \text{Tor } M,$$

such that (i) $F \cong R^s$ for some non-negative integer s , and (ii) $\text{Tor } M \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$, where the a_i 's are non-zero non-unit elements in R such that $a_1 \mid a_2 \mid \cdots \mid a_r$.

Proof. By Theorem 9.1, we have $M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_r$, where the a_i 's are non-zero non-unit elements in R such that $a_1 \mid a_2 \mid \cdots \mid a_r$. It then follows that

$$M = F \oplus T,$$

where $F \cong R^s$, and $T \cong R/Ra_1 \oplus \cdots \oplus R/Ra_r$.

Clearly $a_r T = 0$. Thus $T \subseteq \text{Tor } M$. Next let $x \in \text{Tor } M$. Then $x = x_1 + x_2$, with $x_1 \in F$ and $x_2 \in T$. So $x_1 = x - x_2 \in \text{Tor } M$, and hence $rx_1 = 0$ for some $0 \neq r \in R$. If $(y_1, \dots, y_s) \in R^s$ is the image of x_1 under the isomorphism $F \cong R^s$, then $r(y_1, \dots, y_s) = 0$. Hence $ry_i = 0$ for $1 \leq i \leq s$. Therefore, each $y_i = 0$, because $r \neq 0$. This yields $x_1 = 0$ and proves that $x \in T$, and thus $T = \text{Tor } M$. $\square$

We are almost ready to prove the uniqueness of a decomposition in Theorem 9.1. Recall first from Theorem 6.21 that in a ring with unity, maximal ideals are prime ideals. The converse holds for PIDs:

Theorem 9.8. *Let R be a principal ideal domain. Then any non-zero ideal $P \neq R$ is prime if and only if it is maximal.*

Proof. Let $P = (a)$ be a non-zero prime ideal in R . If P is not maximal, then there exists an ideal M such that $P \subsetneq M$ and $M \neq R$. Now $M = (b)$ for some $b \in R$. Thus $(a) \subseteq (b)$ implies $a = bx$ for some $x \in R$ and $b \notin (a)$. Since $P = (a)$ is prime and $b \notin (a)$, we have $x \in (a)$. Then $x = ay$ for some $y \in R$. Then $a = bx = bay$ implies that $1 = by$, as R is an integral domain. Therefore $M = R$, a contradiction.

The converse is Theorem 6.21. $\square$

Hence the quotient of a PID by a non-zero prime ideal is a field. We will utilise this in the proof of uniqueness below.

Theorem 9.9. *Let R be a principal ideal domain, and let M be a finitely generated R -module. Suppose*

$$(9.1) \quad M \cong R^s \oplus R/Ra_1 \oplus \cdots \oplus R/Ra_u,$$

where the a_i 's are non-zero non-unit elements in R such that $a_1 \mid a_2 \mid \cdots \mid a_u$, and suppose

$$(9.2) \quad M \cong R^t \oplus R/Rb_1 \oplus \cdots \oplus R/Rb_v,$$

where the b_i 's are non-zero non-unit elements in R such that $b_1 \mid b_2 \mid \cdots \mid b_v$. Then $s = t$, $u = v$ and $Ra_i = Rb_i$ for $1 \leq i \leq u$.

Proof. By Theorem 9.7 the equations (9.1) and (9.2) respectively give $M = F \oplus \text{Tor } M$ and $M = F' \oplus \text{Tor } M$, where $F \cong R^s$ and $F' \cong R^t$. Now then

$$\frac{M}{\text{Tor } M} = \frac{F \oplus \text{Tor } M}{\text{Tor } M} \cong F.$$

Thus $F \cong F'$. This yields $s = t$ (cf. Theorem 5.21).

We now proceed to show that if

$$(9.3) \quad T = \text{Tor } M \cong R/Ra_1 \oplus \cdots \oplus R/Ra_u$$

and

$$(9.4) \quad T = \text{Tor } M \cong R/Rb_1 \oplus \cdots \oplus R/Rb_v$$

where the a_i 's and b_i 's satisfy the conditions as in (9.1) and (9.2) respectively, then $u = v$ and $Ra_i = Rb_i$. For any R -module X and a prime $p \in R$, we define

$$X_p = \{x \in X \mid px = 0\}.$$

Then from (9.3),

$$T_p \cong \bigoplus_{i=1}^u \left(\frac{R}{Ra_i} \right)_p.$$

Clearly,

$$\left(\frac{R}{Ra_i} \right)_p = \begin{cases} \frac{Rd_i}{Ra_i} & \text{if } p \mid a_i \text{ and } a_i = pd_i, \\ (0) & \text{if } p \nmid a_i, \end{cases}$$

and Rd_i/Ra_i is a one-dimensional vector space over the field R/Rp generated by the element $d_i + Ra_i$. Therefore T_p is a vector space over R/Rp , and its dimension is equal to the number of terms R/Ra_i such that $p \mid a_i$.

Suppose that p is a prime dividing a_1 , and hence a_i , for each $i = 1, \dots, u$. Let T have a direct sum decomposition into v summands as in (9.4). Then p must divide at least u of the elements b_j , whence $u \leq v$. By symmetry, we have $u = v$. Therefore, any two decompositions of types (9.1) and (9.2) of any given module must have an equal number of summands.

From (9.3) we have $a_u T = 0$. Then (9.4) gives $Ra_u \subseteq Rb_u$. By symmetry $Rb_u \subseteq Ra_u$. Hence $Ra_u = Rb_u$.

Now assume $(a_i) = (b_i)$ for $k \leq i \leq u$. We show $(a_{k-1}) = (b_{k-1})$. This is done by contradiction, where the contradiction is obtained by multiplying (9.3) and (9.4) by a suitable element in R to make the first m summands zero in one of the decompositions (9.3) and (9.4), and the first $k-1$ summands zero in the other decomposition, for some $m < k-1$.

(The remainder of the proof is non-examinable for the oral exam.) Let p be a prime in R with $p^\alpha \mid a_{k-1}$, $p^{\alpha+1} \nmid a_{k-1}$ and $p^\beta \mid b_{k-1}$, $p^{\beta+1} \nmid b_{k-1}$, for some non-zero α, β . If $\alpha = \beta$ for each prime p , then we are done because this means $(a_{k-1}) = (b_{k-1})$. So suppose otherwise, so there is some prime p with corresponding α and β being unequal. For definiteness, let $\alpha > \beta$.

Let θ be such that $p^\theta \mid a_u$, $p^{\theta+1} \nmid a_u$, and write $a_u = b_u = p^\theta a'_u$. Put

$$x = p^{\alpha-1} a'_u \in R.$$

Then (9.3) gives

$$(9.5) \quad xT = x \left(\frac{R}{Ra_1} \oplus \dots \oplus \frac{R}{Ra_u} \right) = \frac{Rx + Ra_1}{Ra_1} \oplus \dots \oplus \frac{Rx + Ra_u}{Ra_u},$$

and its $(k-1)$ st summand is

$$\begin{aligned} \frac{Rx + Ra_{k-1}}{Ra_{k-1}} &= \frac{Rp^{\alpha-1} a'_u + Rp^\alpha a'_{k-1}}{Rp^\alpha a'_{k-1}} \quad (\text{where } (a'_{k-1}, p) = 1) \\ &= \frac{Rd}{Rp^\alpha a'_{k-1}}, \end{aligned}$$

where $d = (p^{\alpha-1} a'_u, p^\alpha a'_{k-1})$, the gcd of $p^{\alpha-1} a'_u$ and $p^\alpha a'_{k-1}$. However

$$\begin{aligned} a_{k-1} \mid a_u &\implies a'_{k-1} \mid a'_u \\ &\implies (p^{\alpha-1} a'_u, p^\alpha a'_{k-1}) = p^{\alpha-1} a'_{k-1}. \end{aligned}$$

Therefore $d = p^{\alpha-1} a'_{k-1}$. Thus the $(k-1)$ st summand in (9.5) is

$$\frac{Rp^{\alpha-1} a'_{k-1}}{Rp^\alpha a'_{k-1}} \cong \frac{R}{Rp},$$

because in any integral domain $Ra/Rab \cong R/Rb$, for $0 \neq a, b \in R$.

It can similarly be shown that any summand preceding the $(k-1)$ st summand is either $R/(p)$ or (0) ; and indeed if any summand is (0) then all the preceding ones are also zero.

Therefore (9.5) can be rewritten as

$$(9.6) \quad xT \cong (0) \oplus \cdots \oplus (0) \oplus R/(p) \oplus \cdots \oplus R/(p) \oplus R/(p^{\lambda_{k-1}}) \oplus \cdots \oplus R/(p^{\lambda_u}),$$

where $s, t \geq 0$ with $s + t = k - 2$ and $\lambda_{k-1}, \dots, \lambda_u \in \mathbb{N}$. Further from (9.4),

$$(9.7) \quad xT = \frac{Rx + Rb_1}{Rb_1} \oplus \cdots \oplus \frac{Rx + Rb_u}{Rb_u},$$

and its $(k-1)$ st summand is

$$\frac{Rx + Rb_{k-1}}{Rb_{k-1}} = \frac{Rp^{\alpha-1}a'_u + Rp^{\beta}b'_{k-1}}{Rp^{\beta}b'_{k-1}} \quad (\text{where } (b'_{k-1}, p) = 1).$$

Since $\alpha > \beta$ and $b_{k-1} \mid b_u = a_u$, we have

$$b'_{k-1} \mid a'_u \implies p^{\beta}b'_{k-1} \mid p^{\alpha-1}a'_u.$$

Therefore

$$\frac{Rx + Rb_{k-1}}{Rb_{k-1}} = \frac{Rp^{\beta}b'_{k-1}}{Rp^{\beta}b'_{k-1}} = (0),$$

Thus $Rx \subseteq Rb_{k-1}$. Since

$$Rb_{k-1} \subseteq Rb_{k-2} \subseteq \cdots \subseteq Rb_1,$$

it follows that the first $k-1$ summands in (9.7) are all zero. Therefore, the decomposition (9.7) of xT may be rewritten as

$$(9.8) \quad xT = (0) \oplus \cdots \oplus (0) \oplus R/(p^{\lambda_k}) \oplus \cdots \oplus R/(p^{\lambda_u}).$$

Comparing (9.6) and (9.8), we obtain a contradiction, because, as remarked earlier, any two such decompositions of a module over a PID must have an equal number of non-zero summands. Hence $\alpha = \beta$. This implies $Ra_{k-1} = Rb_{k-1}$. It then follows that $Ra_i = Rb_i$ for all $i = 1, \dots, u$. $\square$

Note that a different argument is needed to show that $Ra_i = Rb_i$ for $i < u$. This is because a module may have several isomorphic submodules. For example, if $M \cong \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}$, there are two submodules isomorphic to $\mathbb{Z}/2\mathbb{Z}$. The one given by $\mathbb{Z}/2\mathbb{Z} \oplus 0$ and $0 \oplus \mathbb{Z}/2\mathbb{Z}$.

9.3. Application to finitely generated abelian groups. Since $\mathbb{Z}$ is a PID and any abelian group is a $\mathbb{Z}$ -module, an immediate application of Theorems 9.1 and 9.9 gives an alternative proof of the decomposition theorem for a finitely generated abelian group.

Theorem 9.10 (Fundamental Theorem of Abelian Groups). *Let A be a finitely generated abelian group. Then*

$$(9.9) \quad A \cong \mathbb{Z}^s \oplus \mathbb{Z}/a_1\mathbb{Z} \oplus \cdots \oplus \mathbb{Z}/a_r\mathbb{Z},$$

where s is a non-negative integer and the a_i 's are non-zero non-units in $\mathbb{Z}$, such that

$$(9.10) \quad a_1 \mid a_2 \mid \cdots \mid a_r.$$

Further, the decomposition (9.9) of A subject to the condition (9.10) is unique. (Here $\mathbb{Z}^0$ is interpreted as the trivial group (0) .)

In particular, if A is generated by $(x_1, \dots, x_n)$ subject to

$$\sum_{j=1}^n a_{ij}x_j = 0, \quad \text{for } 1 \leq i \leq m,$$

then

$$A \cong \mathbb{Z} \times \overset{n-r}{\dots} \times \mathbb{Z} \times \mathbb{Z}/a_1\mathbb{Z} \times \dots \times \mathbb{Z}/a_r\mathbb{Z}$$

where $a_1, \dots, a_r$ are the invariant factors of the $m \times n$ matrix $A = (a_{ij})$.

Example 9.11. The abelian group generated by x_1 and x_2 subject to $2x_1 = 0$, $3x_2 = 0$ is isomorphic to $\mathbb{Z}/6\mathbb{Z}$, since the matrix (of constraints) is $\begin{pmatrix} 2 & 0 \\ 0 & 3 \end{pmatrix}$, which is equivalent to $\begin{pmatrix} 1 & 0 \\ 0 & 6 \end{pmatrix}$.

Example 9.12. Find the abelian group generated by $\{x_1, x_2, x_3\}$ subject to

$$5x_1 + 9x_2 + 5x_3 = 0$$

$$2x_1 + 4x_2 + 2x_3 = 0$$

$$x_1 + x_2 - 3x_3 = 0.$$

Solution. Let

$$A = \begin{pmatrix} 5 & 9 & 5 \\ 2 & 4 & 2 \\ 1 & 1 & -3 \end{pmatrix}.$$

Perform $C_2 - C_1$, $R_1 - 2R_2$, $C_3 - C_1$, $C_2 - C_1$, $R_3 - R_1$, $(-1)R_3$, $C_1 - C_2$, and $C_2 - C_1$ in succession to obtain that A is equivalent to

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 4 \end{pmatrix}.$$

Hence the desired abelian group is isomorphic to $\mathbb{Z}/1\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z}$.